



COMPARAÇÃO DE LLMs

Comparação de Modelos de Linguagem (LLMs) no Hugging Face

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O objetivo desta atividade é avaliar o desempenho de dois modelos de código aberto — o **gpt2** da OpenAI e o **gpt-neo-125m** da EleutherAI — disponíveis na plataforma Hugging Face. O desafio consiste em testar como modelos de linguagem de pequeno porte lidam com a geração de texto em português, avaliando tanto o **tempo de inferência** quanto a **coerência da resposta**.

Implementação e Metodologia



Ferramentas Utilizadas O Código e o Teste

Utilizamos a classe `pipeline` para instanciar os dois modelos. Ambos foram alimentados com o mesmo *prompt* inicial:

"O futuro da inteligência artificial no Brasil é"

Configuramos a geração para produzir até **50 tokens** de texto, medindo o tempo de execução de cada modelo.

Python

Linguagem principal do projeto

Google Colab

Ambiente de execução na nuvem

Transformers

Biblioteca do Hugging Face

Imagens do código e dos resultados

```
# Instale a biblioteca necessária (rode esta célula primeiro)
!pip install transformers torch

# --- Célula 2: Importação e Configuração ---
from transformers import pipeline
import time

# Definindo o prompt (entrada) para os modelos
prompt = "O futuro da inteligência artificial no Brasil é"

print(f"Prompt de entrada: '{prompt}'\n")
print("-" * 50)

# --- Modelo 1: GPT-2 ---
print("Carregando o Modelo 1: gpt2...")
start_time = time.time()
generator_1 = pipeline('text-generation', model='gpt2')
resultado_1 = generator_1(prompt, max_length=50, num_return_sequences=1, truncation=True)
tempo_1 = time.time() - start_time

print(f"Resultado GPT-2 ((tempo_1:.2f) segundos):")
print(resultado_1[0]['generated_text'])
print("-" * 50)

# --- Modelo 2: GPT-Neo 125M ---
print("Carregando o Modelo 2: EleutherAI/gpt-neo-125m...")
start_time = time.time()
generator_2 = pipeline('text-generation', model='EleutherAI/gpt-neo-125m')
resultado_2 = generator_2(prompt, max_length=50, num_return_sequences=1, truncation=True)
tempo_2 = time.time() - start_time

print(f"Resultado GPT-Neo ((tempo_2:.2f) segundos):")
print(resultado_2[0]['generated_text'])
print("-" * 50)
```

```
... Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.11.0+cpu)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.29.0)
Requirement already satisfied: huggingface-hub<2.0,=>1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.16.1)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.2)
Requirement already satisfied: pyyaml<=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex<=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.22.2)
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers) (0.24.0)
Requirement already satisfied: safetensors>=0.4.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm<4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.4.3 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (1.5.0)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (0.28.1)
Requirement already satisfied: typer<=0.20.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (0.25.1)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2->torch) (3.0.3)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (4.13.0)
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (2026.5.20)
Requirement already satisfied: httpcore<1,* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (1.0.9)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (3.15)
Requirement already satisfied: h11>=0.10 in /usr/local/lib/python3.12/dist-packages (from httpcore<1,*->httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (0.16.0)
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer>=0.20.0->huggingface-hub<2.0,>=1.3.0->transformers) (8.4.0)
Requirement already satisfied: shellingham<1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer>=0.20.0->huggingface-hub<2.0,>=1.3.0->transformers) (1.5.4)
Requirement already satisfied: rich>=13.8.0 in /usr/local/lib/python3.12/dist-packages (from typer>=0.20.0->huggingface-hub<2.0,>=1.3.0->transformers) (13.0.4)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer>=0.20.0->huggingface-hub<2.0,>=1.3.0->transformers) (0.0.4)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=13.8.0->typer>=0.20.0->huggingface-hub<2.0,>=1.3.0->transformers) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=13.8.0->typer>=0.20.0->huggingface-hub<2.0,>=1.3.0->transformers) (2.20.0)
Requirement already satisfied: MarkupSafe<0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=13.8.0->typer>=0.20.0->huggingface-hub<2.0,>=1.3.0->transformers) (0.1.2)
Prompt de entrada: 'O futuro da inteligência artificial no Brasil é'

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Carregando o Modelo 1: gpt2...
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:112: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it as secret in your Google Colab and restart your session.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
config.json: 100% ██████████ 605/605 [00:00<00:00, 45.9kB/s]
model.safetensors: 100% ██████████ 548M/548M [00:05<00:00, 110MB/s]
Loading weights: 100% ██████████ 149/148 [00:00<00:00, 480.020%/s, Materializing param=transformer.wte.weight]
GPT2LMHeadModel LOAD REPORT from: gpt2
Key | Status | |
-----+-----+-----+
h.{0...11}.attn.bias | UNEXPECTED | |

Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
generation_config.json: 100% ██████████ 124/124 [00:00<00:00, 5.01kB/s]
tokenizer_config.json: 100% ██████████ 26.0/26.0 [00:00<00:00, 922B/s]
vocab.json: 100% ██████████ 1.04M/1.04M [00:00<00:00, 5.88MB/s]
merges.txt: 100% ██████████ 456k/456k [00:00<00:00, 3.40MB/s]
tokenizer.json: 100% ██████████ 1.30M/1.30M [00:00<00:00, 0.22MB/s]
Passing 'generation_config' together with generation-related arguments ({'num_return_sequences', 'max_length'}) is deprecated and will be removed in future versions. Please pass either a 'generation_config' object OR all generation parameters explicitly, but not both.
Setting 'pad_token_id' to 'eos_token_id':50256 for open-end generation.
Both 'max_new_tokens' (=256) and 'max_length' (=50) seem to have been set. 'max_new_tokens' will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)
Resultado GPT-2 (25.61 segundos):
O futuro da inteligência artificial no Brasil éque a llega.

D. The meaning of the term "torture" is to punish. The sentence that is used is not to be punished, but to be punished. In Greek the word istos, meaning "to make a man more miserable than he is good, or worse than he is good, to make him worse than he is good, or worse than he is good."

E. The term "torture" is derived from "torture" in Latin, from "torture" in Greek.

F. The word "torture" is used in the same way as the word "torture": namely, it refers to the punishment of the person who is in danger of being killed, the death of a friend, the death of a lover, the death of a brother, the death of a friend's wife, the death of a friend's mother, the death of a friend

G. The term "torture" is used in the same way as the word "torture" in Latin, from "torture" in Greek.

H. The punishment
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Carregando o Modelo 2: EleutherAI/gpt-neo-125m...
config.json: 100% ██████████ 1.01k/1.01k [00:00<00:00, 67.5kB/s]
model.safetensors: 100% ██████████ 520M/520M [00:02<00:00, 235MB/s]
Loading weights: 100% ██████████ 160/160 [00:00<00:00, 434.850%/s, Materializing param=transformer.wte.weight]
GPTNeoForCausalLM LOAD REPORT from: EleutherAI/gpt-neo-125m
Key | Status | |
-----+-----+-----+
transformer.h.{0...11}.attn.attention.masked_bias | UNEXPECTED | |
transformer.h.{0, 2, 4, 6, 8, 10}.attn.attention.bias | UNEXPECTED | |

Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
generation_config.json: 100% ██████████ 119/119 [00:00<00:00, 5.08kB/s]
tokenizer_config.json: 100% ██████████ 727/727 [00:00<00:00, 33.5kB/s]
vocab.json: 100% ██████████ 890k/890k [00:00<00:00, 20.4MB/s]

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F. The word "torture" is used in the same way as the word "torture": namely, it refers to the punishment of the person who is in danger of being killed, the death of a friend, the death of a lover, the death of a brother, the death of a friend's wife, the death of a friend's mother, the death of a friend

G. The term "torture" is used in the same way as the word "torture" in Latin, from "torture" in Greek.

H. The punishment
-----
Carregando o Modelo 2: EleutherAI/gpt-neo-125m...
config.json: 100% ██████████ 1.01k/1.01k [00:00<00:00, 67.5kB/s]
model.safetensors: 100% ██████████ 520M/520M [00:02<00:00, 235MB/s]
Loading weights: 100% ██████████ 160/160 [00:00<00:00, 434.850%/s, Materializing param=transformer.wte.weight]
GPTNeoForCausalLM LOAD REPORT from: EleutherAI/gpt-neo-125m
Key | Status | |
-----+-----+-----+
transformer.h.{0...11}.attn.attention.masked_bias | UNEXPECTED | |
transformer.h.{0, 2, 4, 6, 8, 10}.attn.attention.bias | UNEXPECTED | |

Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
generation_config.json: 100% ██████████ 119/119 [00:00<00:00, 5.08kB/s]
tokenizer_config.json: 100% ██████████ 727/727 [00:00<00:00, 33.5kB/s]
vocab.json: 100% ██████████ 890k/890k [00:00<00:00, 20.4MB/s]
merges.txt: 100% ██████████ 456k/456k [00:00<00:00, 20.7MB/s]
special_tokens_map.json: 100% ██████████ 357/357 [00:00<00:00, 29.2kB/s]
tokenizer.json: 100% ██████████ 2.11M/2.11M [00:00<00:00, 52.0MB/s]
Setting 'pad_token_id' to 'eos_token_id':50256 for open-end generation.
Both 'max_new_tokens' (=256) and 'max_length' (=50) seem to have been set. 'max_new_tokens' will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)
Resultado GPT-Neo (21.82 segundos):
O futuro da inteligência artificial no Brasil é um bom plano de segurança como a sua base de base de dados, a primeira vez que o Brasil está em conta de que a população brasileira está em conta de que ela é aplicada a uma base de dados que ajusta-se. (Foto: Facebook)

Apesar de que o Brasil não vence para o Brasil, o Brasil é um estado de pertencimento à sociedade. Há muito que temos para a sociedade. A inteligência é válida, já que a população brasileira está em conta de que a população brasileira é aplicada de uma base de dados que ajusta-se. (Foto: Facebook)

Ao mesmo tempo, o Brasil está em conta de que o Brasil está em conta de que o Brasil está em conta de que ela é aplicada a uma base de dados que ajusta-se. (Foto:
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```


Análise dos Resultados: GPT-2 vs. GPT-Neo

21.82s

GPT-Neo

Tempo de execução — mais rápido

25.61s

GPT-2

Tempo de execução — mais lento

Qualidade do Texto Gerado

1

GPT-2

Demonstrou alta instabilidade com o português. Após palavras confusas (*"équipe a llega"*), alucinou completamente e mudou para o inglês, gerando texto sem sentido sobre o termo *"torture"*.

2

GPT-Neo

Manteve o idioma em português, mas entrou em **loop de repetição**, gerando frases como *"base de base de dados"* e *"o Brasil está em conta de que..."*.



- ❑ **Conclusão:** Modelos base pequenos (não instruídos/fine-tuned) e treinados predominantemente em inglês sofrem com **alucinações e repetições** ao gerar textos em português. Para aplicações reais no Brasil, seria necessário usar **modelos maiores** ou fazer um *fine-tuning* com datasets em PT-BR.